Mathematical Contribution of SLDE-AFT

1. Scope and interpretation

This section formalizes the confidence aggregation rule used by the probabilistic knowledge base (PKB) variant of SLDE-AFT. The rule combines repeated extraction observations for the same knowledge triple, discounts potentially correlated evidence, and applies a conflict adjustment when incompatible objects are retained for a functionally single-valued subject–predicate slot.

The output is called an **aggregated confidence score** or **probability-like support score**. It is not assumed to be a calibrated posterior probability. A posterior interpretation would require calibrated observation probabilities and a defensible conditional-independence model. Calibration is therefore evaluated empirically rather than assumed.

The mathematical analysis does not replace the original SLDE-AFT architecture. It analyzes the existing PKB module while preserving the dual-source extractor, feedback controller, provenance layer, synthetic-data generator, and automated LoRA adaptation loop.

2. Notation and aggregation rule

Let a knowledge triple be

$$t = (s, p, o),$$

where s, p, and o denote its subject, predicate, and object. Let

$$\mathcal{E}_t = \{c_1, c_2, \dots, c_{n_t}\}, \qquad c_i \in [0, 1],$$

be the confidence values of the n_t extraction observations supporting t. Let $\lambda \in (0, 1]$ be the shrinkage factor. The current implementation uses

$$\lambda = 0.75.$$

For a subject–predicate pair (s, p), define the set of currently retained objects as

$$\Omega_{s,p} = \{o' : (s, p, o') \in \mathrm{PKB}\}.$$

For a triple $t = (s, p, o)$, the number of other competing objects is

$$m_t = |\Omega_{s,p} \setminus \{o\}|.$$

For predicates that are not designated as functionally single-valued, set $m_t = 0$. The Noisy-Or support score is

$$A_t = 1 - \prod_{i=1}^{n_t} (1 - \lambda c_i). \tag{1}$$

For a functionally single-valued predicate, the final conflict-adjusted confidence is

$$C_t = \frac{A_t}{m_t + 1} = \frac{1 - \prod_{i=1}^{n_t}(1 - \lambda c_i)}{m_t + 1}. \tag{2}$$

Equation (1) aggregates repeated evidence for one triple. Equation (2) applies a conservative consistency adjustment when incompatible alternatives exist for the same subject–predicate slot.

The mutual-exclusivity adjustment must be enabled only for predicates for which the single-value assumption is valid. It should not be automatically applied to genuinely multi-valued relations.

3. Formal derivation

3.1 Residual-support model

The effective support contributed by observation i is λc_i. Its residual non-support quantity is therefore

$$1 - \lambda c_i.$$

Under the conditional-independence approximation, the joint residual support after all observations is

$$R_t = \prod_{i=1}^{n_t}(1 - \lambda c_i).$$

The complement gives the support that at least one observation supports the triple:

$$A_t = 1 - R_t = 1 - \prod_{i=1}^{n_t}(1 - \lambda c_i).$$

The shrinkage factor does not establish independence. It reduces the contribution of each observation to account for possible correlation, such as repeated extraction from similar documents, overlapping evidence, or repeated use of the same model and prompt.

3.2 Conflict adjustment

Suppose several candidate triples share the same subject and predicate but have different objects. For a functionally single-valued predicate, these candidates are competing explanations. The factor

$$\frac{1}{m_t + 1}$$

reduces the score of each candidate when other candidates are retained. The factor is a consistency regularizer; it is not a normalized probability distribution over the alternatives.

Thus, Equation (2) should be interpreted as a conservative ranking and admission score for PKB use, especially for deciding whether a triple is eligible for synthetic supervision.

4. Theoretical properties

4.1 Lemma 1: evidence monotonicity

Lemma 1. Fix a triple t, shrinkage factor $\lambda \in (0, 1]$, and competitor count m_t. Let $C_t^{(n)}$ be the score after n supporting observations. For any additional observation $c_{n+1} \in [0, 1]$,

$$C_t^{(n+1)} \geq C_t^{(n)}.$$

Proof. Define

$$R_n = \prod_{i=1}^{n} (1 - \lambda c_i).$$

Since $c_{n+1} \in [0, 1]$ and $\lambda \in (0, 1]$,

$$0 \leq 1 - \lambda c_{n+1} \leq 1.$$

Therefore,

$$R_{n+1} = R_n(1 - \lambda c_{n+1}) \leq R_n.$$

Consequently,

$$A_t^{(n+1)} = 1 - R_{n+1} \geq 1 - R_n = A_t^{(n)}.$$

Because $m_t + 1$ is fixed and positive,

$$C_t^{(n+1)} = \quad racA_t^{(n+1)} m_t + 1 \geq \frac{A_t^{(n)}}{m_t + 1} = C_t^{(n)}.$$

□

This result guarantees monotonicity of the Noisy-Or support when the conflict set is fixed. It does not guarantee monotonicity of the final score when new competing alternatives are discovered.

4.2 Dynamic-conflict limitation

If a new competing object is added, then

$$m_t^{(n+1)} > m_t^{(n)}.$$

Although evidence monotonicity still gives

$$A_t^{(n+1)} \geq A_t^{(n)},$$

it is possible that

$$\frac{A_t^{(n+1)}}{m_t^{(n+1)}+1} < \frac{A_t^{(n)}}{m_t^{(n)}+1}.$$

This decrease is intentional: the system has discovered additional conflict. Accordingly, SLDE-AFT provides two separate guarantees:

- **Retention monotonicity:** an admitted PKB record is not deleted by the accumulation procedure.
- **Evidence monotonicity:** the Noisy-Or support for a fixed triple does not decrease when additional supporting observations are added.

The final conflict-adjusted confidence is not guaranteed to be non-decreasing when the competitor set changes.

4.3 Proposition 1: boundedness

Proposition 1. For any $n_t \geq 0$, $c_i \in [0,1]$, $\lambda \in (0,1]$, and $m_t \geq 0$,

$$0 \leq C_t \leq \frac{1}{m_t+1} \leq 1.$$

Proof. Each factor $(1-\lambda c_i)$ lies in $[0,1]$, so their product lies in $[0,1]$. Therefore $A_t \in [0,1]$. Dividing by $m_t + 1 \geq 1$ gives

$$0 \leq C_t \leq \frac{1}{m_t+1} \leq 1.$$

□

4.4 Proposition 2: convergence under a fixed conflict set

Proposition 2. Assume that m_t remains fixed and that there exist $\epsilon > 0$ and infinitely many observations satisfying $c_i \geq \epsilon$. Then

$$\lim_{n\to\infty} A_t^{(n)} = 1$$

and

$$\lim_{n\to\infty} C_t^{(n)} = \frac{1}{m_t+1}.$$

Proof sketch. For every observation satisfying $c_i \geq \epsilon$,

$$1-\lambda c_i \leq 1-\lambda\epsilon < 1.$$

The product of infinitely many such factors converges to zero. Therefore $R_t \to 0$, which implies $A_t \to 1$. Since m_t is fixed, Equation (2) gives

$$C_t \to \frac{1}{m_t+1}.$$

□

This is a convergence guarantee for the aggregation operator under a stable competitor set. It is not a guarantee that the extracted triple is factually correct.

5. Computational complexity

5.1 Current prototype

The current prototype stores the complete confidence history for each triple and recomputes the residual product when a duplicate observation is received. If triple t has n_t observations, this update costs

$$O(n_t).$$

The prototype identifies competing objects by scanning retained PKB entries for the same subject–predicate pair. If the PKB contains $|\mathrm{KB}|$ entries, this lookup costs up to

$$O(|\mathrm{KB}|).$$

For a candidate batch $\mathcal{C}$, a worst-case merge therefore costs

$$O\left(\sum_{t\in\mathcal{C}} n_t + |\mathcal{C}|\,|\mathrm{KB}|\right). \tag{3}$$

If all confidence histories and provenance records are retained, memory usage is

$$O\left(|\mathrm{KB}| + \sum_{t} n_t\right), \tag{4}$$

excluding the variable length of provenance text.

5.2 Optimized implementation

For scalability experiments, maintain the residual product directly:

$$R_t \leftarrow R_t(1-\lambda c), \qquad A_t \leftarrow 1 - R_t. \tag{5}$$

Also maintain a hash index

$$(s, p) \mapsto \Omega_{s,p}.$$

With these structures, duplicate aggregation is expected $O(1)$, conflict lookup is expected $O(1)$, and processing a candidate batch of size $|\mathcal{C}|$ is expected $O(|\mathcal{C}|)$, assuming constant-time hash operations. The total memory remains linear in the number of retained triples and stored observations.

The optimized complexity must be reported only if the indexed running-residual implementation is actually used in the experiment. Otherwise, report the prototype complexity from Equation (3).

6. Confidence calibration analysis

The theoretical properties above describe the behavior of the score function. They do not prove that a score of 0.90 corresponds to a 90% probability of correctness. Calibration must therefore be measured against gold labels.

For every retained triple t, define

$$y_t = \begin{cases} 1, & \text{if } t \text{ matches a gold triple,} \\ 0, & \text{otherwise.} \end{cases}$$

Partition the scores into non-overlapping bins $B_1, \ldots, B_K$. For each non-empty bin, calculate empirical accuracy:

$$\mathrm{acc}(B_k) = \frac{1}{|B_k|} \sum_{t \in B_k} y_t, \tag{6}$$

and mean predicted confidence:

$$\mathrm{conf}(B_k) = \frac{1}{|B_k|} \sum_{t \in B_k} C_t. \tag{7}$$

The Expected Calibration Error is

$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{|B_k|}{N} |\mathrm{acc}(B_k) - \mathrm{conf}(B_k)|, \tag{8}$$

where N is the total number of evaluated triples. Lower ECE indicates closer agreement between confidence and empirical correctness.

The paper must report a table such as:

Confidence bin	Count	Mean confidence	Empirical accuracy	Calibration gap
[0.00, 0.80)	to measure	to measure	to measure	to measure
[0.80, 0.85)	to measure	to measure	to measure	to measure
[0.85, 0.90)	to measure	to measure	to measure	to measure
[0.90, 0.95)	to measure	to measure	to measure	to measure
[0.95, 1.00]	to measure	to measure	to measure	to measure

A reliability diagram should plot mean confidence on the x-axis and empirical accuracy on the y-axis. The diagonal $y = x$ represents perfect calibration.

Because $\tau = 0.88$ is used as the synthetic-data admission threshold, evaluate the threshold directly. Report the number of retained triples, precision, recall, and F1 for

$$\tau \in \{0.80, 0.85, 0.88, 0.90, 0.95\}.$$

The threshold should be selected using a validation split or development set rather than tuned on the final test set.

7. Empirical comparison of aggregation rules

To demonstrate that the probabilistic component contributes beyond the general SLDE-AFT pipeline, compare aggregation rules using identical extraction outputs, datasets, thresholds, and random seeds:

1. **Max merge:**
$$C_t = max_i c_i.$$
2. **Mean aggregation:**
$$C_t = frac1n_t sum\_i = 1^{n_t} c_i.$$
3. **Standard Noisy-Or:**
$$A_t = 1 - prod_i(1 - c_i).$$
4. **Conservative Noisy-Or:**
$$A_t = 1 - prod_i(1 - 0.75c_i).$$
5. **Conservative Noisy-Or with conflict adjustment:**
$$C_t = frac1 - prod_i(1 - 0.75c_i)m_t + 1.$$

For each method, report:

- precision;
- recall;
- F1;
- ECE;
- retained-triple count;
- conflict count;
- runtime;
- synthetic-example count, if the rule is used to generate training data.

This experiment is necessary to show whether the shrinkage factor and conflict adjustment provide measurable value rather than merely adding mathematical notation.

8. Empirical convergence experiment

For each extraction iteration r, record:

$$A_t^{(r)}, \qquad C_t^{(r)}, \qquad m_t^{(r)},$$

for every tracked triple. Report:

- mean Noisy-Or support by iteration;
- mean final confidence by iteration;
- number of retained triples;

- number of newly discovered conflicts;
- number of triples crossing the synthetic-data threshold.

Plot the mean and distribution of $A_t^{(r)}$ and $C_t^{(r)}$ over iterations. This provides empirical evidence about convergence behavior while respecting the theoretical limitation that dynamic conflicts can reduce the final adjusted score.

9. Validity and limitations

The mathematical guarantees apply to the aggregation operator, not to the correctness of the underlying extracted facts. In particular:

- repeated correlated hallucinations may still produce high support;
- the shrinkage parameter reduces but does not eliminate correlation effects;
- the conflict penalty is a heuristic regularizer, not a normalized posterior over alternatives;
- calibration must be measured empirically;
- functional-predicate assumptions must be explicitly documented;
- threshold selection must not use the final test set.

The PKB therefore combines mathematical aggregation with provenance checks, source-diversity checks, benchmark evaluation, confidence calibration, and error analysis before using triples as synthetic supervision.

10. Integration with the original SLDE-AFT design

The revised mathematical contribution preserves the original SLDE-AFT workflow:

$$\text{dual-source ingestion} \rightarrow \text{extraction} \rightarrow \text{PKB accumulation} \rightarrow \text{feedback} \rightarrow \text{synthetic supervisior}$$

The contribution of the mathematical module is the explicit and testable treatment of confidence accumulation within the PKB. The final paper should claim that the method provides bounded and evidence-monotone support under stated assumptions, not that it guarantees factual truth or universal calibration.

Once the calibration table, reliability diagram, threshold sweep, aggregation ablation, and convergence plot are populated with actual experimental results, this section will address the requested formal derivation, theoretical justification, convergence discussion, complexity analysis, confidence calibration, and probabilistic-accumulation guarantees while retaining the original SLDE-AFT architecture.